Splitting methods for rare event simulations
Part 2: The dynamic case

T. Lelièvre

MODAL Math App SNA

3 avril 2026

Multilevel splitting

General setting: Let $(\boldsymbol{X}_t)_{t\geq 0}$ be a Markovian dynamics, and τ_B and τ_A two associated stopping times.

Objective: efficiently compute quantities of the form $\mathbb{E}[F((\boldsymbol{X}_t)_{0\leq t\leq \tau_A\wedge\tau_B})1_{\tau_B<\tau_A}]$ in the rare event setting:

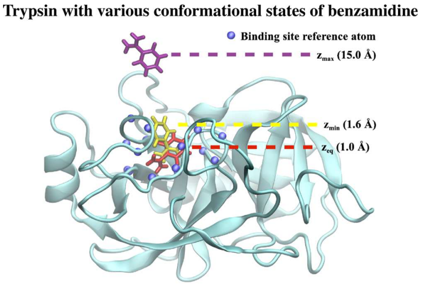
$$\mathbb{P}(\tau_B < \tau_A) \ll 1.$$

Two examples:

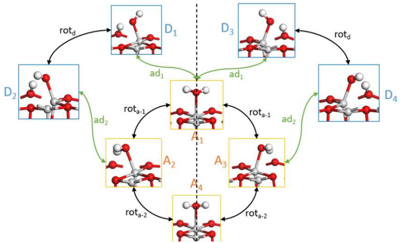
- Reactive trajectories: A and B are two metastable states, τ_A and τ_B are the first hitting time of A and B.
- Killed process: τ_A is a killing time, τ_B is the first hitting time of a domain B.

Motivation 1: Simulations of molecular systems

Unbinding of a ligand from a protein



Catalytic reaction: dissociation of water

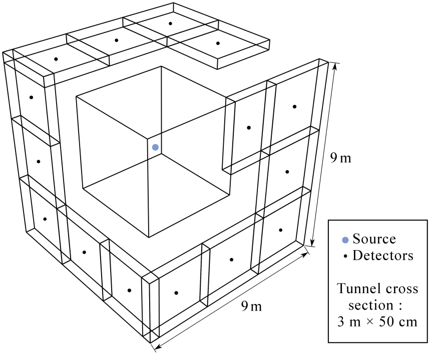


Elementary time-step for the molecular dynamics $= 10^{-15}\,\mathrm{s}$
Macroscopic timescales of interest $\simeq 10^{-9} - 10^{-2}\,\mathrm{s}$

Challenge: bridge the gap between timescales

Motivation 2: Radiation protection

Monte Carlo particle transport



Concrete tunnel with a neutron source

How to compute the neutron flux at the detector ?

Challenge: the flux is very small

Multilevel splitting: the reactive trajectory setting

We would like to sample trajectories between two given metastable states A and B. The main assumption is that we are given a smooth one dimensional function $\xi : \mathbb{R}^d \to \mathbb{R}$ which "indexes" the transition from A to B in the following sense:

$$A \subset \{x \in \mathbb{R}^d, \xi(x) < z_{\min}\} \text{ and } B \subset \{x \in \mathbb{R}^d, \xi(x) > z_{\max}\},$$

where $z_{\min} < z_{\max}$, and $\Sigma_{z_{\min}}$ (resp. $\Sigma_{z_{\max}}$) is "close" to ∂A (resp. ∂B).

Example: $\xi(x) = \|x - x_A\|$ where x_A is a reference configuration in A.

We are interested in the event $\{\tau_B < \tau_A\}$, starting from an initial condition with support in $\{x \in \mathbb{R}^d, \xi(x) < z_{\min}\}$, where

$$\tau_A = \inf\{t > 0, \boldsymbol{X}_t \in A\}, \quad \tau_B = \inf\{t > 0, \boldsymbol{X}_t \in B\}.$$

Multilevel splitting

Objective: Simulate efficiently trajectories which reach B before A and estimate $\mathbb{P}(\tau_B < \tau_A)$. This then gives dynamical information: reactive trajectories from A to B, transition times from A to B, ...

We present a multilevel splitting approach [Kahn, Harris, 1951] [Rosenbluth, 1955] to discard failed trajectories and branch trajectories approaching the rare set. We focus on an adaptive variant [Cérou, Guyader, 2007] [Cérou, Guyader, TL, Pommier, 2010]: the Adaptive Multilevel Splitting (AMS) algorithm.

Remark: The algorithm can be seen as a kind of adaptive Forward Flux Sampling [Allen, Valeriani, Ten Wolde, 2009]. It is also related to the Interface Sampling Method [Bolhuis, van Erp, Moroni 2003] and the Milestoning method [Elber, Faradjian 2004]. See the review paper [Bolhuis, Dellago, 2009]

Reactive trajectory

A reactive trajectory between two metastable sets A and B is a piece of equilibrium trajectory that leaves A and goes to B without going back to A in the meantime [Hummer,2004] [Metzner, Schütte, Vanden-Eijnden, 2006].

Difficulty: A trajectory leaving A is more likely to go back to A than to reach B.

Splitting algorithm: basic idea

The idea of splitting algorithms (FFS, RESTART, ...) is to write the rare event

$$\{\tau_B < \tau_A\}$$

as a sequence of nested events: for $z_{\min} = z_1 < \ldots < z_Q = z_{\max}$,

$$\{\tau_{z_1} < \tau_A\} \supset \{\tau_{z_2} < \tau_A\} \supset \ldots \supset \{\tau_{z_{\max}} < \tau_A\} \supset \{\tau_B < \tau_A\}$$

where $\tau_z = \inf\{t > 0, \xi(\boldsymbol{X}_t) > z\}$ and to simulate the successive *conditional events*: for $q = 1, \ldots, Q-1$,

$$\{\tau_{z_{q+1}} < \tau_A\} \text{ knowing that } \{\tau_{z_q} < \tau_A\}.$$

It is then easy to build an unbiased estimator of

$$\mathbb{P}(\tau_B < \tau_A) = \mathbb{P}(\tau_{z_1} < \tau_A)\mathbb{P}(\tau_{z_2} < \tau_A|\tau_{z_1} < \tau_A)\ldots\mathbb{P}(\tau_B < \tau_A|\tau_{z_{\max}} < \tau_A)$$

Splitting algorithm: adaptive level computation

Problem: How to choose the intermediate levels $(z_q)_{q\geq 1}$?

In an ideal setting, for a given number of intermediate levels, the optimum in terms of variance is attained if

$$\forall q \geq 1,\ \mathbb{P}(\tau_{z_q} < \tau_A | \tau_{z_{q-1}} < \tau_A) = \mathbb{P}(\tau_{z_2} < \tau_A | \tau_{z_1} < \tau_A).$$

This naturally leads to an adaptive version (AMS, nested sampling) where the levels are determined by using *empirical quantiles*: Fix $k < n$; at iteration $q \geq 1$, given n trajectories $(\boldsymbol{X}^{\ell}_{t\wedge\tau_A})_{t>0,\ell=1,\ldots,n}$ in the event $\{\tau_{z_{q-1}} < \tau_A\}$, choose z_q so that

$$\mathbb{P}(\tau_{z_q} < \tau_A | \tau_{z_{q-1}} < \tau_A) \simeq \left(1 - \frac{k}{n}\right).$$

The level z_q is the k-th order statistics of $\sup_{t\geq 0} \xi(\boldsymbol{X}^{\ell}_{t\wedge\tau_A})$:

$$\sup_{t\geq 0} \xi(\boldsymbol{X}^{(1)}_{t\wedge\tau_A}) < \ldots < \sup_{t\geq 0} \xi(\boldsymbol{X}^{(k)}_{t\wedge\tau_A}) =: z_q < \ldots < \sup_{t\geq 0} \xi(\boldsymbol{X}^{(n)}_{t\wedge\tau_A}).$$

AMS: estimator of the rare event probability (1/2)

Let Q_{iter} be the number of iterations to reach the level $z_{\max}$:

$$Q_{\mathrm{iter}} = \min\{q \geq 0, z_q > z_{\max}\}$$

(where z_0 is the k-th order statistics of the n initial trajectories). Then, one obtains the estimator:

$$\left(1 - \frac{k}{n}\right)^{Q_{\mathrm{iter}}} \simeq \mathbb{P}(\tau_{z_{\max}} < \tau_A).$$

AMS: estimator of the rare event probability (2/2)

At iteration $Q_{\rm iter}$, one has an ensemble of n trajectories such that $\tau_{z_{\max}} < \tau_A$. Thus

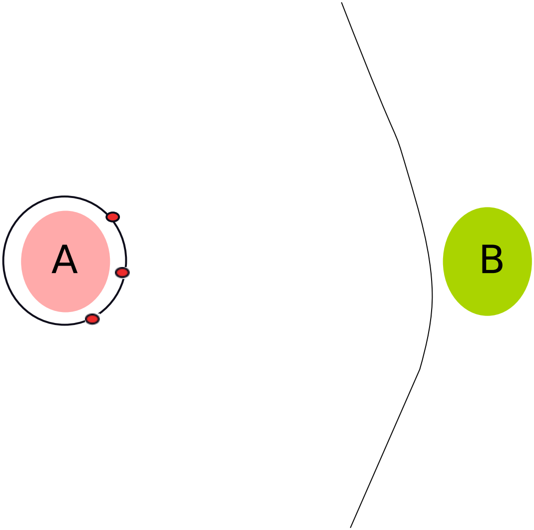
$$\hat{p}_{\rm corr} := \frac{1}{n}\sum_{\ell=1}^{n} 1_{\{T_B(\boldsymbol{X}^{\ell,Q_{\rm iter}})<T_A(\boldsymbol{X}^{\ell,Q_{\rm iter}})\}} \simeq \mathbb{P}(\tau_B < \tau_A | \tau_{z_{\max}} < \tau_A).$$

$\hat{p}_{\rm corr}$ is the number of trajectories reaching B before A at the last iteration $Q_{\rm iter}$.

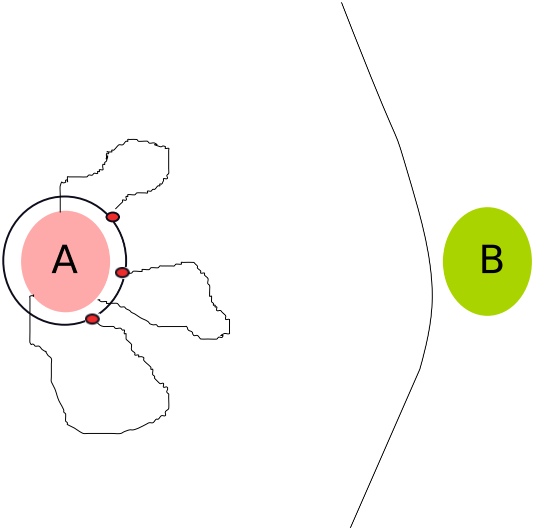
Therefore, an estimator of $\mathbb{P}(\tau_B < \tau_A)$ is

$$\left(1-\frac{k}{n}\right)^{Q_{\rm iter}} \hat{p}_{\rm corr}.$$

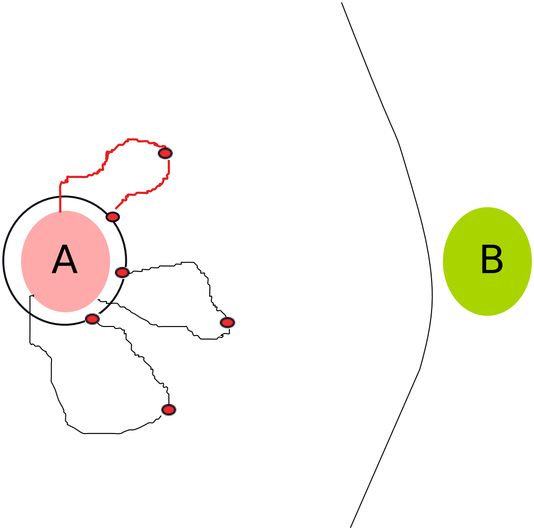
AMS Algorithm



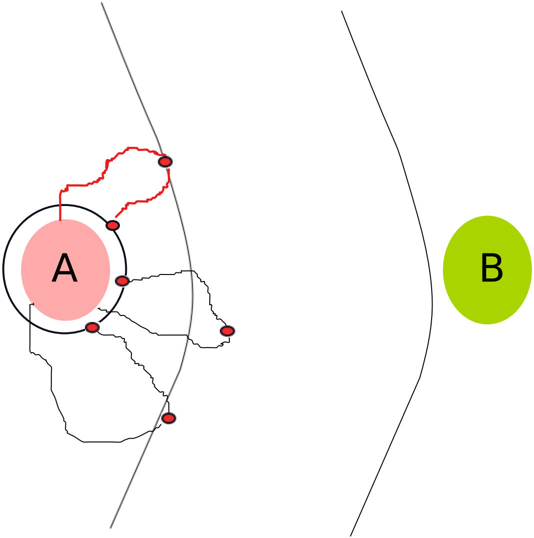
AMS Algorithm



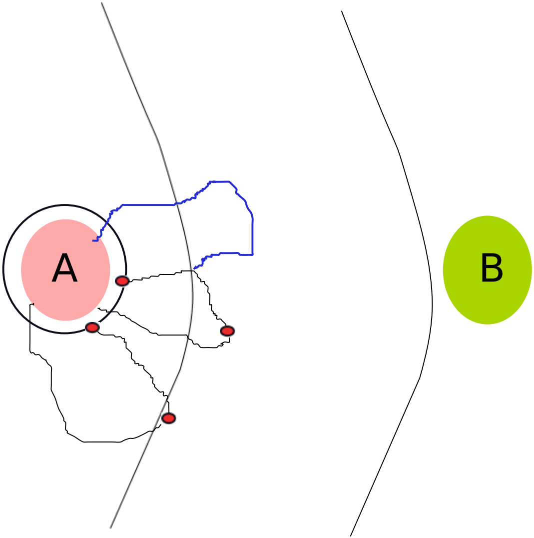
AMS Algorithm



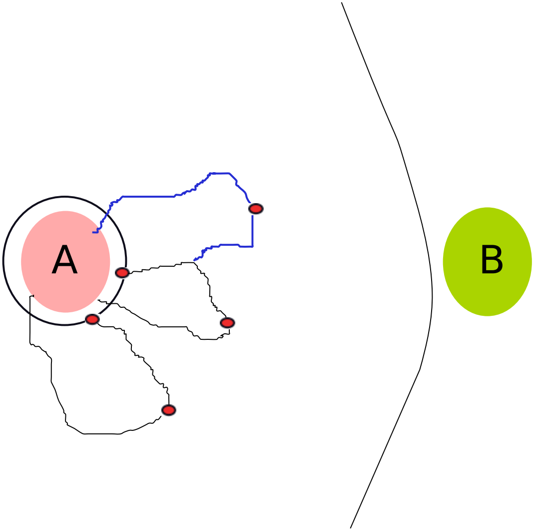
AMS Algorithm



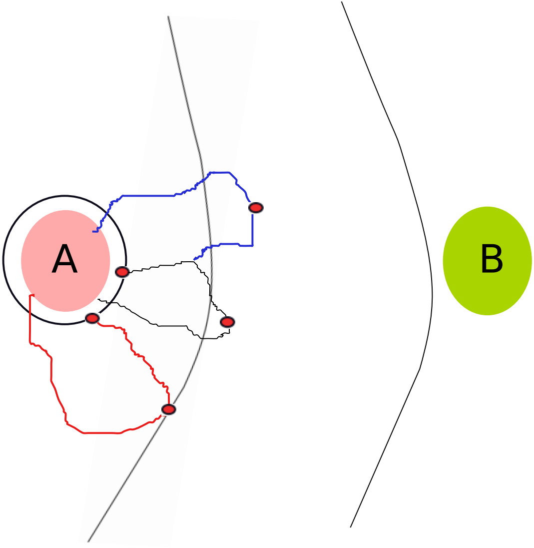
AMS Algorithm



AMS Algorithm



AMS Algorithm



AMS Algorithm: the case of Markov chains

In practice, the dynamics are *discrete in time* and thus, it may happen that more than k trajectories are such that

$$\sup_{t \geq 0} \xi(\boldsymbol{X}^{\ell}_{t \wedge \tau_A}) \leq \sup_{t \geq 0} \xi(\boldsymbol{X}^{(k)}_{t \wedge \tau_A}) =: z_q$$

In this case, all the trajectories with maximum level smaller or equal than z_q should be discarded.

The actual estimator of $\mathbb{P}(\tau_B < \tau_A)$ thus reads:

$$\hat{p} = \left(1 - \frac{K_1}{n}\right) \dots \left(1 - \frac{K_{Q_{\mathrm{iter}}}}{n}\right) \hat{p}_{\mathrm{corr}}$$

instead of $\left(1 - \frac{k}{n}\right)^{Q_{\mathrm{iter}}} \hat{p}_{\mathrm{corr}}$, where $K_q \geq k$ is the effective number of discarded trajectories at iteration q.

AMS Algorithm: unbiasedness

Theorem [C.-E. Bréhier, M. Gazeau, L. Goudenège, TL, M. Rousset, 2016]: For any choice of ξ, n and k,

$$\mathbb{E}(\hat{p}) = \mathbb{P}(\tau_B < \tau_A).$$

The proof is based on Doob's stopping theorem on a martingale built using filtrations indexed by the level sets of ξ. Actually, this result is proved for general path observables and in a much more general setting.

Practical counterparts:

- The algorithm is easy to parallelize.
- One can compare the results obtained with different score functions ξ to gain confidence in the results.

AMS Algorithm: dynamics versus static

Comparison with the static version presented in part 1:

- The resampling is very easy to do, and does not require many iterations to get a new sample
- One obtains an unbiased estimator whatever the parameters ξ, n and k.

It is possible to get approximations of conditional expectations:

$$\mathbb{E}[F((\boldsymbol{X}_t)_{0\le t\le \tau_A\wedge\tau_B})|\tau_B<\tau_A] = \frac{\mathbb{E}[F((\boldsymbol{X}_t)_{0\le t\le \tau_A\wedge\tau_B})1_{\tau_B<\tau_A}]}{\mathbb{P}(\tau_B<\tau_A)}.$$

AMS yields unbiased estimators of the numerator and denominator

(but thus a biased estimator of the ratio, the bias goes to zero when the number of AMS runs goes to infinity).

In practice, the choice of the score function ξ is crucial!

In the hands-on session, we will consider a 1d case for which the choice of ξ is not a problem.

On the choice of the score function

It can be shown at least in some specific settings [Cérou, Delyon, Guyader, Rousset, 2019] that the optimal choice of the score function in terms of variance is the so-called committor function:

$$\xi^*(\boldsymbol{x}) = \mathbb{P}_{\boldsymbol{x}}(\tau_B < \tau_A).$$

In this case, $\sqrt{n}(\hat{p}_n - p)$ converges (when $n \to \infty$) in distribution towards a centered Gaussian random variable with variance $-p^2 \ln p$.

Remarks:

- Knowing ξ^* is essentially equivalent to knowing p...
- Actually, any score function with the same level sets as ξ^* leads to the optimal variance.
- This has been proven for continuous-in-time diffusion processes.

Computing transition times: the Hill relation

To use the algorithm to compute transition times, we split a transition path from A to B into: excursions from ∂A to $\Sigma_{z_{\min}}$ and then back to ∂A, and finally an excursion from ∂A to $\Sigma_{z_{\min}}$ and then to B. Assuming that A is metastable ($p \ll 1$), it can be shown that the equilibrium mean transition time can be approximated by:

$$\left(\frac{1}{p} - 1\right) \Delta_{\mathrm{Loop}} + \Delta_{React}$$

where:

- p is the probability, starting from $\Sigma_{z_{\min}}$ "at equilibrium", to go to B rather than A (approximated by $\hat{p}$) ;
- Δ_{Loop} is the mean time for an excursion from ∂A to $\Sigma_{z_{\min}}$ and then back to ∂A (approximated by brute force) ;
- Δ_{React} is the mean time for an excursion from ∂A to $\Sigma_{z_{\min}}$ and then to B (approximated by the AMS algorithm).

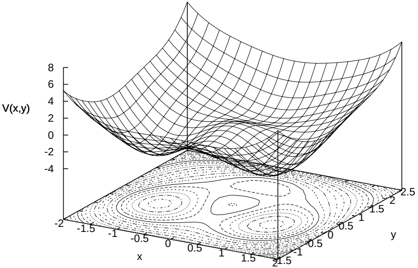
Numerical results: a 2D example

Time-discretization of the overdamped Langevin dynamics:

$$d\boldsymbol{X}_t = -\nabla V(\boldsymbol{X}_t)\,dt + \sqrt{2\beta^{-1}}d\boldsymbol{W}_t$$

with a deterministic initial condition $\boldsymbol{X}_0 = \boldsymbol{x}_0$ and the 2D potential

[Park, Sener, Lu, Schulten, 2003] [Metzner, Schütte and Vanden-Eijnden, 2006]



$$V(x,y) = 3\mathrm{e}^{-x^2-(y-\frac{1}{3})^2} - 3\mathrm{e}^{-x^2-(y-\frac{5}{3})^2} - 5\mathrm{e}^{-(x-1)^2-y^2}$$

$$- 5\mathrm{e}^{-(x+1)^2-y^2} + 0.2x^4 + 0.2\left(y-\frac{1}{3}\right)^4.$$

A 2D example

The interest of this “bi-channel” potential is that, depending on the temperature, one or the other channel is prefered to go from A (around $H_- = (-1,0)$) to B (around $H_+ = (1,0)$).

Three score functions: $\xi^1(x,y) = \|(x,y) - H_-\|$, $\xi^2(x,y) = C - \|(x,y) - H_+\|$ or $\xi^3(x,y) = x$.

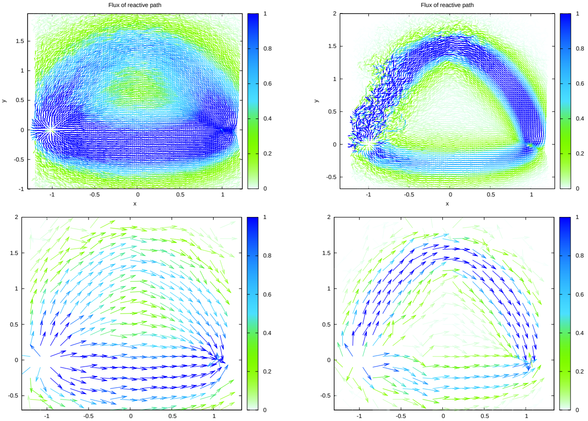
We plot as a function of the number N of independent realizations of AMS, the empirical average

$$\overline{p}_N = \frac{1}{N} \sum_{m=1}^{N} \hat{p}_m$$

together with the associated empirical confidence interval: $[\overline{p}_N - \delta_N/2, \overline{p}_N + \delta_N/2]$ where

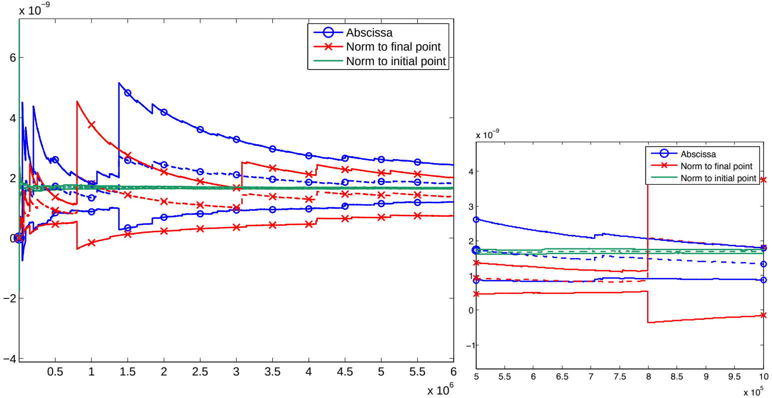
$$\delta_N = 2\frac{1.96}{\sqrt{N}} \sqrt{\frac{1}{N} \sum_{m=1}^{N} (\hat{p}_m)^2 - (\overline{p}_N)^2}$$

A 2D example: flux of reactive trajectories

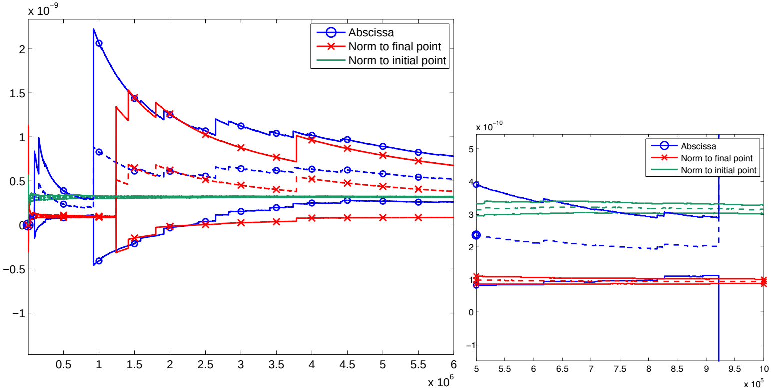


Flux of reactive trajectories, at $\beta = 1.67$ on the left, and $\beta = 6.67$ on the right.

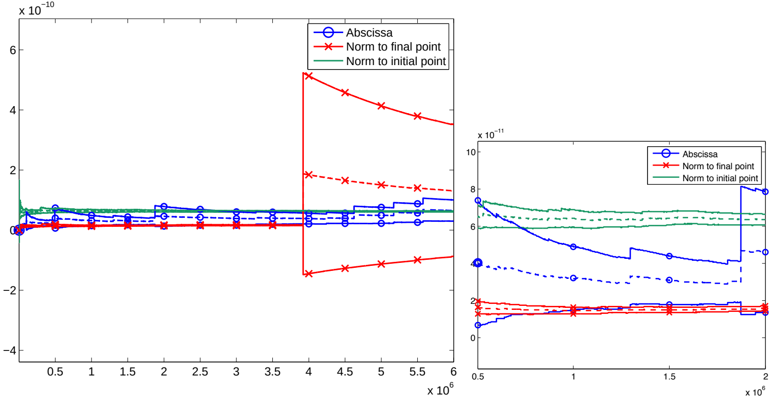
A 2D example: $k = 1$, $n = 100$, $\beta = 8.67$



A 2D example: $k = 1$, $n = 100$, $\beta = 9.33$



A 2D example: $k = 1$, $n = 100$, $\beta = 10$



A 2D example

Observations:

- When N is sufficiently large, confidence intervals overlap.
- For too small values of N, "apparent bias" is observed [Glasserman, Heidelberger, Shahabuddin, Zajic, 1998].
- Fluctuations depend a lot on ξ.

$\longrightarrow$ To gain confidence in the results, check that the estimated quantity is approximately the same for different ξ's.

"Apparent bias" phenomenon

The apparent bias is due to the fact that [Glasserman, Heidelberger, Shahabuddin, Zajic, 1998]:

- Multiple pathways exist to go from A to B.
- Conditionally to reach Σ_z before A, the relative likelihood of each of these pathways depends a lot on z.

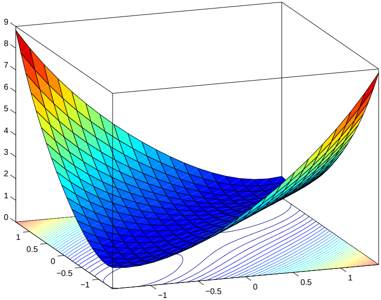
On our example, for small n, we indeed observe that (for ξ^3):

- Most of the time, all replicas at the end go through only one of the two channels (two possible scenarios).
- One of this scenario is rare.
- The values of $\hat{p}$ associated to each of these two scenarios are very different.

This explains the large fluctuations.

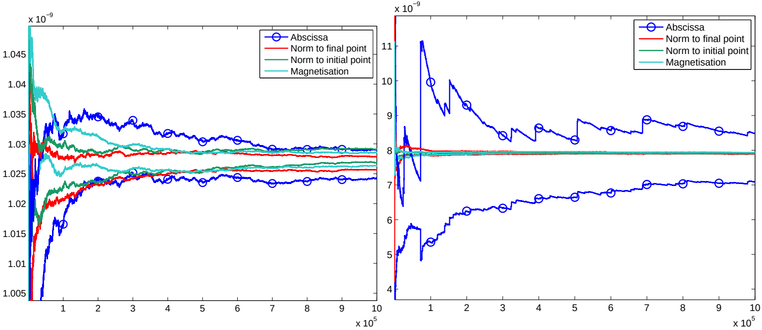
“Apparent bias” phenomenon

Another 2D test case:



Potential $V_\gamma(x,y)$.
Left: $\gamma = 1$ (one channel); right: $\gamma = 0.1$ (two channels).

"Apparent bias" phenomenon



Parameters: $k = 1$, $n = 100$ and $\beta = 80$.
Left: $\gamma = 1$ (one channel). Right: $\gamma = 0.1$ (two channels).

Results on larger test cases

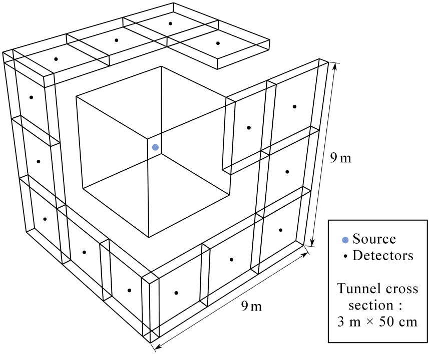
AMS has been implemented in the NAMD software (collaboration with SANOFI, C. Mayne and I. Teo, PhD of L. Silva Lopes with J. Hénin).

AMS has been implemented in Tripoli (collaboration with CEA, C. Diop, E. Dumonteil, PhD of H. Louvin).

AMS is currently tested for Ab Initio Molecular Dynamics with IFPEN (catalytic reaction, collaboration with IFPEN, PhD of T. Pigeon and P. Marmey).

Radiation protection (1/2)

Monte Carlo particle transport

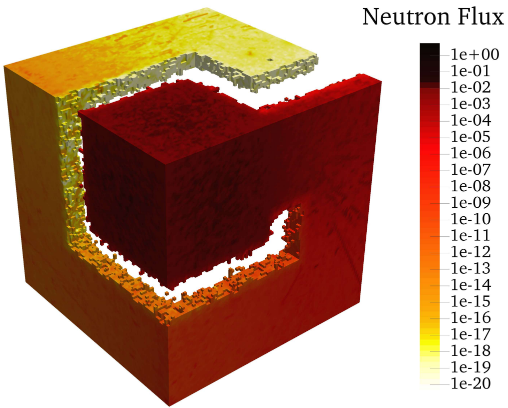


Concrete tunnel with a neutron source

How to compute the neutron flux at the detector ?

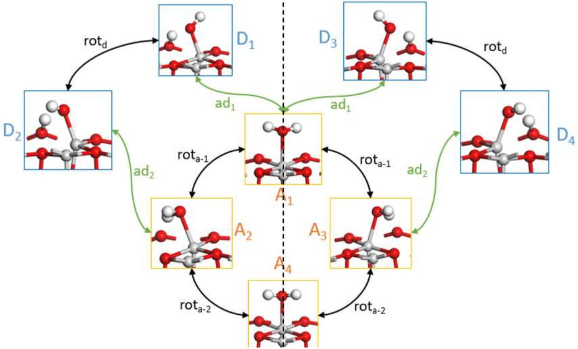
Challenge: the flux is very small

Radiation protection (2/2)



Dissociation of water on (100) surface (1/3)

Metastable states of H_2O on the (100) surface of γ-alumina



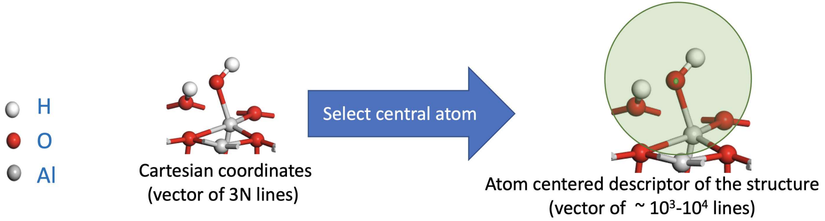
This is a multistate problem. AMS can be used to analyze the transitions:

- Start from A_1, choose $R = A_1$ and $P = A_2A_3 \cup A_4 \cup D_1D_3 \cup D_2D_4$: What is the most probable exit from A_1?
- Start from A_1, choose $R = A_1 \cup A_2A_3 \cup A_4 \cup D_2D_4$ and $P = D_1D_3$: Focus on the $A_1 \to D_1D_3$ transition

Construction of the SOAP-SVM collective variable (2/3)

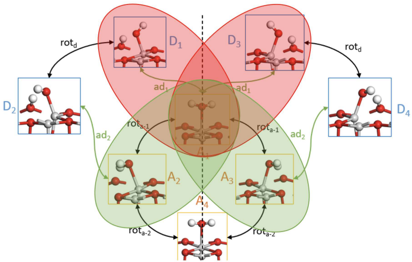
Two ingredients to build the score function:

- Smooth Overlap of Atomic Positions (SOAP) atom centered descriptors [Bartok, Kondor, Csanyi, Phys. Rev. B, 2013]
- Support-Vector Machine (SVM) to build a classifier decision function to separate the states (using short dynamics in each state)



Comparison with harmonic Transition State Theory (3/3)

Here are the results obtained on two specific transitions (dissociation and rotation)
[$\sim 2.\,10^6$ CPU hours for the whole reaction network]



Dissociation	Hill	hTST
$k_{A_1 \to D_1 D_3}$	$1.6\,10^9 s^{-1}$	$3.4\,10^{11} s^{-1}$
$k_{D_1 D_3 \to A_1}$	$2.3\,10^{10} s^{-1}$	$1.1\,10^{12} s^{-1}$

Rotation	Hill	hTST
$k_{A_1 \to A_2 A_3}$	$3.8\,10^{10} s^{-1}$	$7.6\,10^{10} s^{-1}$
$k_{A_2 A_3 \to A_1}$	$1.5\,10^{11} s^{-1}$	$2.1\,10^{12} s^{-1}$

hTST rates are larger (entropy estimation, recrossing)

Concluding remarks on AMS (1/2)

Practical recommendations:

- A careful implementation of the splitting step leads to unbiased estimators for non-normalized quantities.
- Perform many independent realizations of AMS.
- Use ξ as a numerical parameter.

The algorithm is very versatile:

- Non-intrusivity: the MD integrator is a black box.
- Can be adapted to generate trajectories of any stopped process.
- Can be applied to both entropic and energetic barriers, to non-equilibrium systems, non-homogeneous Markov process, random fields, ...
- Algorithmic variants: other resampling procedure, additional selection, ...

Concluding remarks on AMS (2/2)

Works in progress:

- Adaptive computation of better and better ξ.
- Analysis of the efficiency as a function of ξ. For optimal choice of ξ, the cost of AMS is (for n large)

$$\left((\ln p)^2 - \ln p\right)$$

much better than the cost of naive Monte Carlo: $\frac{1-p}{p}$. How does this degrade when ξ departs from the optimal case ?

References

- T. Lelièvre, M. Ramil and J. Reygner, *Estimation of statistics of transitions and Hill relation for Langevin dynamics*, Ann. IHP 2024.
- C.-E. Bréhier, M. Gazeau, L. Goudenège , TL and M. Rousset, *Unbiasedness of some generalized Adaptive Multilevel Splitting algorithms*, Ann. App. Prob., 2016.
- F. Cérou, B. Delyon, A. Guyader, and M. Rousset, *On the asymptotic normality of Adaptive Multilevel Splitting*, JUQ, 2019.
- F. Cérou, A. Guyader, TL and D. Pommier, *A multiple replica approach to simulate reactive trajectories*, J. Chem. Phys. 2011.
- C.M. Diop, E. Dumonteil, TL, H. Louvin and M. Rousset, *Adaptive Multilevel Splitting for Monte Carlo particle transport*, Eur. Phys. J. N, 2017.
- T. Pigeon, G. Stoltz, M. Corral-Valero, A. Anciaux-Sedrakian, M. Moreaud, TL, P. Raybaud, *Computing Surface Reaction Rates by Adaptive Multilevel Splitting Combined with Machine Learning and Ab Initio Molecular Dynamics*, JCTC 2023